

Homework 5

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May 20, 2026

Problem 1:

(1) There are three types of roots:

- $e_i - e_j = \sum_{k=i}^{j-1} \alpha_k, 1 \leq i < j \leq n;$
- $2e_i = 2 \sum_{k=i}^{n-1} \alpha_k + \alpha_n;$
- $e_i + e_j = \sum_{k=i}^{j-1} \alpha_k + 2 \sum_{k=j}^{n-1} \alpha_k + \alpha_n, 1 \leq i < j \leq n;$

Together with their negatives form the root system of C_n .

(2) Three types of Weyl group elements are corresponding to the three types of roots (x is an arbitrary vector $x = x_i e_i$):

- $s_{e_i - e_j} x = (x_1, \dots, x_{i-1}, x_j, x_{i+1}, \dots, x_{j-1}, x_i, x_{j+1}, \dots, x_n)$, making the reflection on $\mathbb{R}^n$ by swapping the i -th and j -th coordinates;
- $s_{2e_i} x = (x_1, \dots, x_{i-1}, -x_i, x_{i+1}, \dots, x_n)$, making the reflection on $\mathbb{R}^n$ by changing the sign of the i -th coordinate;
- $s_{e_i + e_j} x = (x_1, \dots, x_{i-1}, -x_j, x_{i+1}, \dots, x_{j-1}, -x_i, x_{j+1}, \dots, x_n)$, making the reflection on $\mathbb{R}^n$ by swapping the i -th and j -th coordinates and changing the signs of both coordinates;

We can choose the first two reflections to generate the whole Weyl group.

(3) Choose the simple roots of B_n :

$$\beta_1 = e_1 - e_2, \beta_2 = e_2 - e_3, \dots, \beta_{n-1} = e_{n-1} - e_n, \beta_n = e_n,$$

while the coroots are

$$H_{\beta_1} = e_1 - e_2 = \alpha_1, H_{\beta_2} = e_2 - e_3 = \alpha_2, \dots, H_{\beta_n} = 2e_n = 2\alpha_n.$$

WE find the corresponding Weyl group are the same as C_n , where $s_{e_i} = s_{2e_i}$, so the only thing left to check is

$$\phi(s_\alpha \cdot e_i) = s_{\phi\alpha} \cdot (\phi e_i).$$

Define the linear transformation $\phi : \beta_i \rightarrow \alpha_i$, then under the Weyl group action $s_\alpha \cdot e_i = \pm e_i$ or 0. Thus

$$\phi(s_\alpha \cdot e_i) = \begin{cases} \phi e_i = 2e_i \\ \phi(-e_i) = -2e_i \\ \phi 0 = 0 \end{cases} = s_{\phi\alpha} \cdot (\phi e_i).$$

(4) Symmetrically, the extra white node must be $\alpha_0 = -2e_1$ and it is easy to verify the connectivity of the extended Dynkin diagram. Thus apparently it is a non-simple root of C_n .

Problem 2:

The Chevalley basis of $\mathfrak{sl}(3, \mathbb{C})$ is

$$\begin{aligned} H_{\alpha_1} &= \alpha_1 = E_{11} - E_{22}, H_{\alpha_2} = \alpha_2 = E_{22} - E_{33}, \\ E_{\alpha_1} &= E_{12}, E_{\alpha_2} = E_{23}, E_{\alpha_1+\alpha_2} = E_{13}, \\ E_{-\alpha_1} &= E_{21}, E_{-\alpha_2} = E_{32}, E_{-(\alpha_1+\alpha_2)} = E_{31}. \end{aligned}$$

It is easy to verify the Lie brackets of the Chevalley basis.

Problem 3:

(1) First consider simple Lie algebra $\mathfrak{g}$ with rank $\mathfrak{g} = 3$:

- A_3 : $\bullet \text{---} \bullet \text{---} \bullet$;
- B_3 : $\bullet \text{---} \bullet \rightleftarrows \bullet$;
- C_3 : $\bullet \text{---} \bullet \rightleftarrows \bullet$;
- D_3 : $\bullet \text{---} \bullet \text{---} \bullet \cong A_3$;

So non-isomorphic rank-3 simple Lie algebras are A_3 , B_3 and C_3 . Similarly we find non-isomorphic rank-2 simple Lie algebras A_2 , B_2 and G_2 , and non-isomorphic rank-1 simple Lie algebra A_1 . Then we can construct semi-simple Lie algebra using direct sum:

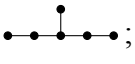
- $A_2 \oplus A_1$;
- $B_2 \oplus A_1$;
- $G_2 \oplus A_1$;
- $A_1 \oplus A_1 \oplus A_1$.

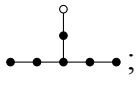
(2) The dimension of the semi-simple Lie algebra is the sum of the numbers of roots and the rank, so we have

- $\dim(A_3) = 15$
- $\dim(B_3) = 21$
- $\dim(C_3) = 21$
- $\dim(A_2 \oplus A_1) = 11$;
- $\dim(B_2 \oplus A_1) = 13$;
- $\dim(G_2 \oplus A_1) = 17$;
- $\dim(A_1 \oplus A_1 \oplus A_1) = 9$.

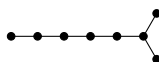
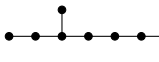
Problem 4:

(1) no. $\mathfrak{su}(10) \cong A_9$ and $\text{rank}(A_9) > \text{rank}(E_8)$;

(2) yes. $\mathfrak{so}(10) \cong D_5$:  $\Rightarrow D_5 \oplus \mathfrak{u}(1) \subset E_6$: ;

(3) no. $\mathfrak{so}(12) \cong D_6$: , $\hat{E}_6$: ;

(4) no. $\dim(F_4) = 52 > \dim(A_6) = 48$;

(5) yes. $\mathfrak{so}(16) \cong D_8$: , $\hat{E}_8$: , just delete the first left node;

(6) yes. $\mathfrak{su}(4) \cong A_3$, $\mathfrak{su}(2) \cong A_1$, $\mathfrak{so}(10) \cong D_5$, $\hat{D}_5$: , just delete one of the middle nodes to get $A_3 \oplus A_1 \oplus A_1$;

(7) no. $\mathfrak{su}(4) \cong A_3$, $\mathfrak{su}(3) \cong A_2$, $\mathfrak{so}(10) \cong D_5$, $\hat{D}_5$: , we can't obtain $A_3 \oplus A_2$ by deleting one node in $\hat{D}_5$;

(8) yes. C_3 : $\bullet \rightleftarrows \bullet$ is apparently a subdiagram of F_4 : $\bullet \rightleftarrows \bullet \rightleftarrows \bullet$.

(9) yes. Just delete one node in both A_3 and A_4 and we can get A_3 and A_2 , together with $\mathfrak{u}(1)^{\oplus 2}$ form a subalgebra of $A_3 \oplus A_4$.

Problem 5:

(1) The generators of $\mathfrak{so}(4)$ are $\{J_{ij}^{(4)} = E_{ij} - E_{ji} \mid 1 \leq i < j \leq 4\}$, while the generators of $\mathfrak{so}(5)$ are $\{J_{ij}^{(5)} = E_{ij} - E_{ji} \mid 1 \leq i < j \leq 5\}$. Thus we can form $\tilde{J}_{ij}^{(5)} = \begin{pmatrix} J_{ij}^{(4)} & 0 \\ 0 & 0 \end{pmatrix}$, which are the generators of $\mathfrak{so}(5)$ and also forms a closed vector space. So $\mathfrak{so}(4)$ is a subalgebra of $\mathfrak{so}(5)$.

(2) Consider the Cartan subalgebra of $\mathfrak{so}(4)$ as the 2×2 block-diagonal matrices. If we identify $\mathfrak{h}$ with $\mathbb{C}^2$ by means of the map

$$H \mapsto i(a_1, a_2),$$

then the roots are $\pm e_1 \pm e_2$, so we can write the root system as $\{\pm \alpha_1, \pm \alpha_2\}$. This can also be derived from $\mathfrak{so}(4) \cong D_2 \cong A_1 \oplus A_1$.

For $\mathfrak{so}(5) \cong B_2$, the root system is $\{\pm e_1, \pm e_2, \pm e_1 \pm e_2\}$.

(3) $\pm e_1 \pm e_2$ are the long roots of B_2 , and the correspondence ensures the closure of the root system.